The Store Model as a Set of Axioms

A human-readable account of the Lean 4 proofs and the Alloy checks

remote-store, BK-435 (exposition)

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Abstract

A store is a rooted tree of folders and items, each addressed by one Key. Spec 051 states that tree as eight axioms, and states the consequences a caller relies on: one entity per Key, the shape of a Key, and closure of Keys under ancestors. This note proves these in ordinary mathematical prose, following the Lean 4 formalization. It also proves the formalization's further result: up to isomorphism, the models are the non-empty prefix-closed sets of name sequences in which the items are marked, every item a non-empty leaf. No result proved in Lean assumes the store is finite. A last section explains what the bounded Alloy checks add.

Status of this document. This is an *exposition*, not a source of truth. The authority for the statements is spec 051, and the machine-checked authority for the proofs is the Lean file. Each result proved in Lean names, after its heading, the Lean declaration it restates. One result, Proposition 6.3, is checked only in Alloy and says so. Where this text and the Lean file disagree, the Lean file is right and this text is wrong. The proofs follow the Lean proofs step by step, with the bookkeeping that a proof assistant needs (index arithmetic, coercions, choice of representatives) told in words.

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1 Notation and the axioms

Words. Fix an alphabet Σ and a distinguished symbol $/ \in \Sigma$, the *separator*. Σ^* is the set of finite words over Σ, and ε is the empty word. Words are compared exactly, symbol by symbol. Juxtaposition uv denotes the concatenation of words. The *names* are

$$\mathsf{N} \;=\; \{\, w \in \Sigma^* : w \neq \varepsilon \text{ and } / \text{ does not occur in } w \,\}.$$

Spec 051 asks Σ to contain at least one symbol besides /. The proofs below need that only in Theorem 6.1; elsewhere Σ may be $\{/\}$, in which case N is empty.

Sequences. A finite sequence of names is written $\langle w_1, \ldots, w_n \rangle$, the empty sequence is $\langle\rangle$, $s \frown t$ is the concatenation of sequences, $|s|$ is the length, and $s{\restriction}i$ is the prefix of s of length i (for $i \leq |s|$). Sequences of names and words are different things: a sequence of names is a list of words, and the bridge between them is the join of Definition 2.1.

Structures. A *structure* is a tuple $(U, F, I, r, C, \mathrm{name})$ where U is any set (the universe of *entities*), $F, I \subseteq U$ (*folders* and *items*), $r \in U$ (the *root*), $C \subseteq U \times U$ is a binary relation (*containment*; $p\ C\ x$ reads "p contains x"), and $\mathrm{name}\colon U \to \Sigma^*$. Nothing is assumed about the size of U. [Lean: `Structure`]

Definition 1.1 (Chain)**.** [Lean: `IsChain`] For $x \in U$ and $n \in \mathbb{N}$, a *chain to x of length n* is a sequence $x_0, x_1, \ldots, x_n$ of entities with $x_0 = r$, $x_n = x$, and $x_i\ C\ x_{i+1}$ for every $i < n$.

The axioms. A *model* is a structure that satisfies the following. The axiom numbers are spec 051's; TREE-00k is AX-k. [Lean: `Axioms`]

Axiom AX-1 (partition)**.** $U = F \cup I$ and $F \cap I = \emptyset$.

Axiom AX-2 (root)**.** $r \in F$.

Axiom AX-3 (items are leaves)**.** $C \subseteq F \times U$: if $p\ C\ x$ then $p \in F$.

Axiom AX-4 (at most one container)**.** If $p\ C\ x$ and $q\ C\ x$ then $p = q$.

Axiom AX-5 (root is topmost)**.** There is no p with $p\ C\ r$.

Axiom AX-6 (reachability)**.** Every $x \in U$ has a chain (of some length $n \geq 0$).

Axiom AX-7 (names)**.** $\mathrm{name}(r) = \varepsilon$, and $\mathrm{name}(x) \in \mathsf{N}$ for every $x \neq r$.

Axiom AX-8 (sibling names)**.** If $p\ C\ x$, $p\ C\ y$ and $\mathrm{name}(x) = \mathrm{name}(y)$, then $x = y$.

AX-1 and AX-7 are conjunctions. The Lean file has one field per clause: AX-1 *disjoint* and *covering*, and AX-7 *root*, *non-empty* and *no /*. Section 7 refutes each clause separately. AX-4 asserts only uniqueness. That every non-root *has* a container is a theorem (D1). No finiteness axiom is included.

2 Words: the join of a sequence of names

Definition 2.1 (Join)**.** [Lean: `join`] For a sequence of words, $\mathrm{join}\langle\rangle = \varepsilon$, $\mathrm{join}\langle w\rangle = w$, and

$$\mathrm{join}\langle w, v, \dots\rangle \;=\; w/\,\mathrm{join}\langle v, \dots\rangle.$$

So $\mathrm{join}\langle w_1, \dots, w_n\rangle = w_1/w_2/\cdots/w_n$.

Everything about Keys that is not about trees is a fact about this function. The central fact is that join loses no information on sequences of names (Lemma 2.4). Its engine is the following observation about where the first separator is.

Lemma 2.2 (First separator)**.** [Lean: `split_first`] *Let $w, a, \rho, J \in \Sigma^*$ with / not occurring in w, and suppose $a/\rho = w/J$. Then either $a = w$ and $\rho = J$, or there is a' with $a = w/a'$ and $a'/\rho = J$.*

Proof. Induction on the length of w. If $w = \varepsilon$, the equation reads $a/\rho = /J$. If $a = \varepsilon$, then $\rho = J$ and the first alternative holds. If $a = b\,a''$ for a symbol b, then comparing first symbols gives $b = /$ and $a''/\rho = J$, so the second alternative holds with $a' = a''$.

If $w = c\,w'$ for a symbol c, then $c \neq /$ and $/ \notin w'$. If $a = \varepsilon$, the equation reads $/\rho = c\,w'/J$, which forces $c = /$, a contradiction. So $a = b\,a''$, and comparing first symbols gives $b = c$ and $a''/\rho = w'/J$. By induction, either $a'' = w'$ and $\rho = J$, giving $a = c\,w' = w$, or $a'' = w'/a'$ with $a'/\rho = J$, giving $a = c\,w'/a' = w/a'$. □

Lemma 2.3. [Lean: `join_eq_nil`] *If every entry of s is in N and $\mathrm{join}\,s = \varepsilon$, then $s = \langle\rangle$.*

Proof. If $s = \langle w, \dots\rangle$ is non-empty, $\mathrm{join}\,s$ begins with w (it is w, or $w/\cdots$), and $w \neq \varepsilon$, so $\mathrm{join}\,s \neq \varepsilon$. □

Lemma 2.4 (Join is injective on names)**.** [Lean: `join_inj`] *If every entry of s and of t is in N and $\mathrm{join}\,s = \mathrm{join}\,t$, then $s = t$.*

Proof. Induction on s, by cases on the shapes of s and t.

- If $s = \langle\rangle$ then $\mathrm{join}\,t = \varepsilon$, so $t = \langle\rangle$ by Lemma 2.3. Symmetrically if $t = \langle\rangle$.
- If $s = \langle w\rangle$ and $t = \langle v\rangle$ then $w = v$.
- If $s = \langle w\rangle$ and $t = \langle v, v', \dots\rangle$, then $w = v/\,\mathrm{join}\langle v', \dots\rangle$ contains /, contradicting $w \in \mathsf{N}$. Symmetrically if s is the longer one.
- If $s = \langle w, w', \dots\rangle$ and $t = \langle v, v', \dots\rangle$, then $v/\,\mathrm{join}\langle v', \dots\rangle = w/\,\mathrm{join}\langle w', \dots\rangle$. Since $/ \notin w$, Lemma 2.2 gives either $v = w$ and $\mathrm{join}\langle v', \dots\rangle = \mathrm{join}\langle w', \dots\rangle$, in which case the induction hypothesis gives $\langle w', \dots\rangle = \langle v', \dots\rangle$ and so $s = t$; or $v = w/a'$, which contains /, contradicting $v \in \mathsf{N}$. □

Definition 2.5 (Good shape)**.** [Lean: `GoodShape`] A word k has *good shape* if it does not begin with /, does not end with /, and does not contain //.

Lemma 2.6. [Lean: `join_shape`] *If $s \neq \langle\rangle$ and every entry of s is in N, then $\mathrm{join}\,s \neq \varepsilon$ and $\mathrm{join}\,s$ has good shape.*

Proof. Induction on s. If $s = \langle w\rangle$, then $\mathrm{join}\,s = w$ is non-empty and contains no / at all, so it has good shape trivially.

If $s = \langle w, v, \dots\rangle$, let $J = \mathrm{join}\langle v, \dots\rangle$. By induction $J \neq \varepsilon$ and J has good shape. The word w/J is non-empty because w is. We use Lemma 2.2 with $/ \notin w$ three times.

- *No leading /.* If $w/J = /b$, that is $\varepsilon/b = w/J$, then either $\varepsilon = w$, which is false, or $\varepsilon = w/a'$, which is impossible.
- *No trailing /.* If $w/J = a/$, that is $a/\varepsilon = w/J$, then either $J = \varepsilon$, which is false, or $J = a'/$ for some a', which contradicts the good shape of J.
- *No //.* If $w/J = a//b$, that is $a/(/b) = w/J$, then either $J = /b$, so J begins with /, or $J = a'//b$. Both contradict the good shape of J. □

3 Chains, depth and the Key

From here to the end of Section 4, fix a model $(U, F, I, r, C, \text{name})$.

Lemma 3.1 (Chain uniqueness)**.** [Lean: `chain_unique`] *If $x_0, \ldots, x_n$ and $y_0, \ldots, y_m$ are chains to the same entity x, then $m = n$ and $y_i = x_i$ for all $i \le n$.*

Proof. Induction on n, for all x, m and y at once.

$n = 0$. Then $x = x_0 = r$. If $m \ge 1$, then $y_{m-1}\ C\ y_m = x = r$, contradicting AX-5. So $m = 0$ and $y_0 = r = x_0$.

$n = k + 1$. Then $x_k\ C\ x$. If $m = 0$, then $x = y_0 = r$ and $x_k\ C\ r$, contradicting AX-5. So $m = j + 1$ and $y_j\ C\ x$. By AX-4, $y_j = x_k$. Now $x_0, \ldots, x_k$ and $y_0, \ldots, y_j$ are both chains to x_k, so the induction hypothesis gives $j = k$ and $y_i = x_i$ for $i \le k$. Hence $m = n$, and also $y_n = x = x_n$. □

Definition 3.2 (Depth and the chain of x)**.** [Lean: `depth, chain`] By AX-6 every x has a chain. By Lemma 3.1 its length and entries do not depend on which chain is chosen. Call the length the *depth* of x, $\text{depth}(x)$, and the chain *the chain of x*, written $x_0, \ldots, x_{\text{depth}(x)}$.

The Lean file picks a chain with the axiom of choice and then shows that any other chain agrees with it (`chain_eq`). The following facts are immediate:

- A prefix $x_0, \ldots, x_i$ of a chain is a chain to x_i (`chain_prefix`).
- $\text{depth}(r) = 0$, since r alone is a chain to r, and conversely $\text{depth}(x) = 0$ implies $x = x_0 = r$ (`depth_root`, `eq_root_of_depth`).
- No entry x_i with $i \ge 1$ of a chain is r, since $x_{i-1}\ C\ x_i$ and AX-5 (`chain_ne_root`).

Lemma 3.3 (One step)**.** [Lean: `chain_step`] *If $p\ C\ x$, then $\text{depth}(x) = \text{depth}(p) + 1$, the chain of x agrees with the chain of p at every index $i \le \text{depth}(p)$, and its last entry is x.*

Proof. Append x to the chain of p. Since $p\ C\ x$, the result is a chain to x of length $\text{depth}(p)+1$. By Lemma 3.1 it is the chain of x. □

Theorem 3.4 (D1: every non-root has exactly one container)**.** [Lean: `D1`] *If $x \ne r$, there is a p with $p\ C\ x$, and every q with $q\ C\ x$ equals p.*

Proof. Since $x \ne r$, $\text{depth}(x) = k + 1$ for some k. In the chain of x, $x_k\ C\ x_{k+1} = x$, so $p = x_k$ contains x. Uniqueness is AX-4. □

Definition 3.5 (Names along a chain; Key)**.** [Lean: `seqOf, Key`] For x with chain $x_0, \ldots, x_n$, let

$$\text{seq}(x) = \langle \text{name}(x_1), \ldots, \text{name}(x_n) \rangle, \qquad \text{Key}(x) = \text{join}\,\text{seq}(x) = \text{name}(x_1)/\cdots/\,\text{name}(x_n).$$

The root's name is not used, and $\text{Key}(r) = \text{join}\langle\rangle = \varepsilon$. This is TREE-009.

By construction $|\mathrm{seq}(x)| = \mathrm{depth}(x)$ and $\mathrm{seq}(r) = \langle\rangle$ (`length_seqOf`, `seqOf_root`). The entries of $\mathrm{seq}(x)$ are names: each x_i with $i \geq 1$ is not r, so $\mathrm{name}(x_i) \in \mathsf{N}$ by AX-7 (`seqOf_InN`).

Lemma 3.6. [Lean: `seqOf_step`] *If $p\ C\ x$, then $\mathrm{seq}(x) = \mathrm{seq}(p) \frown \langle \mathrm{name}(x) \rangle$.*

Proof. By Lemma 3.3, the chain of x is the chain of p followed by x. □

Lemma 3.7. [Lean: `seqOf_chain, seqOf_take`] *For the chain $x_0, \ldots, x_n$ of x and every $i \leq n$, $\mathrm{seq}(x_i) = \mathrm{seq}(x) \restriction i$.*

Proof. $x_0, \ldots, x_i$ is a chain to x_i, hence by Lemma 3.1 the chain of x_i. Its names are the first i entries of $\mathrm{seq}(x)$. □

Lemma 3.8 (seq is injective)**.** [Lean: `seqOf_inj`] *If $\mathrm{seq}(x) = \mathrm{seq}(y)$, then $x = y$.*

Proof. Induction on $n = \mathrm{depth}(x)$. Equal sequences have equal lengths, so $\mathrm{depth}(y) = n$ as well.

If $n = 0$, both x and y are r.

If $n = k + 1$, neither is r. By D1 they have containers $p\ C\ x$ and $q\ C\ y$, and by Lemma 3.6

$$\mathrm{seq}(p) \frown \langle \mathrm{name}(x) \rangle = \mathrm{seq}(q) \frown \langle \mathrm{name}(y) \rangle.$$

Cancelling the last entry gives $\mathrm{seq}(p) = \mathrm{seq}(q)$ and $\mathrm{name}(x) = \mathrm{name}(y)$. Since $\mathrm{depth}(p) = k$ by Lemma 3.3, the induction hypothesis gives $p = q$. Then p contains both x and y, and they have one name, so $x = y$ by AX-8. □

This lemma is where AX-8 does its work. Every injectivity statement below reduces to it.

4 The lemmas L1 to L5

Theorem 4.1 (L1, chain uniqueness)**.** [Lean: `L1`] *Every x has exactly one chain: a chain exists, and any two agree in length and in every entry.*

Proof. Existence is AX-6. Uniqueness is Lemma 3.1. □

Theorem 4.2 (L2, acyclicity)**.** [Lean: `L2`] *There are no $k \geq 1$ and $y_0, \ldots, y_k$ with $y_i\ C\ y_{i+1}$ for all $i < k$ and $y_k = y_0$.*

Proof. By Lemma 3.3, each step of C raises the depth by exactly one. By induction on i, $\mathrm{depth}(y_i) = \mathrm{depth}(y_0) + i$ for every $i \leq k$. If $y_k = y_0$, then $\mathrm{depth}(y_0) = \mathrm{depth}(y_0) + k$, so $k = 0$, a contradiction. □

Remark 4.3. The research document derives L2 differently. It walks backwards from y_0 around the cycle by AX-4, and that walk must be the chain of y_0 and so reach r, which then has a container. The depth argument above is the one the Lean file uses. Both use AX-4, through Lemma 3.1. AX-5 and AX-6 alone do not suffice: the structure $C = \{(r, g_1), (g_1, g_2), (g_2, g_1)\}$ satisfies both and has a cycle.

Theorem 4.4 (L3, Key injectivity; TREE-010)**.** [Lean: `L3`] *If $\mathrm{Key}(x) = \mathrm{Key}(y)$, then $x = y$.*

Proof. $\mathrm{join}\,\mathrm{seq}(x) = \mathrm{join}\,\mathrm{seq}(y)$, and all entries are names, so $\mathrm{seq}(x) = \mathrm{seq}(y)$ by Lemma 2.4, and $x = y$ by Lemma 3.8. □

Theorem 4.5 (L4, Key shape; TREE-011)**.** [Lean: `L4`] *$\mathrm{Key}(x) = \varepsilon$ if and only if $x = r$. For $x \neq r$, $\mathrm{Key}(x)$ has good shape: no leading /, no trailing /, and no //.*

Proof. If $x = r$, $\mathrm{Key}(x) = \mathrm{join}\langle\rangle = \varepsilon$. If $\mathrm{Key}(x) = \varepsilon$, then $\mathrm{seq}(x) = \langle\rangle$ by Lemma 2.3, so $\mathrm{depth}(x) = 0$ and $x = r$. If $x \neq r$, then $\mathrm{seq}(x) \neq \langle\rangle$ (as just shown) and its entries are names, so Lemma 2.6 applies. □

Definition 4.6 (Proper ancestor)**.** [Lean: `ProperAncestor`] a is a *proper ancestor* of x if there are $k \geq 1$ and $a = z_0, z_1, \ldots, z_k = x$ with $z_j \; C \; z_{j+1}$ for every $j < k$.

Theorem 4.7 (L5, prefix closure; TREE-012)**.** [Lean: `L5`]

(a) Let $s_1, \ldots, s_n \in \mathsf{N}$ with $\mathrm{Key}(x) = s_1/\cdots/s_n$. For each $0 < i < n$ there is a folder a that is a proper ancestor of x with $\mathrm{Key}(a) = s_1/\cdots/s_i$.

(b) In any chain $x_0, \ldots, x_n$, no x_i with $i < n$ is an item.

Proof. (a) Let $s = \langle s_1, \ldots, s_n\rangle$. By Lemma 2.4, $\mathrm{seq}(x) = s$, so $\mathrm{depth}(x) = n$. Let $x_0, \ldots, x_n$ be the chain of x and take $a = x_i$. It is a folder by AX-3, since $x_i \; C \; x_{i+1}$. It is a proper ancestor of x, witnessed by $x_i, x_{i+1}, \ldots, x_n$ with $n - i \geq 1$ steps. And by Lemma 3.7, $\mathrm{Key}(x_i) = \mathrm{join}\,\mathrm{seq}(x)\restriction i = \mathrm{join}\langle s_1, \ldots, s_i\rangle$.

(b) If $i < n$ then $x_i \; C \; x_{i+1}$, so $x_i \in F$ by AX-3, hence $x_i \notin I$ by AX-1 (disjoint). □

Remark 4.8. The hypothesis $s_j \in \mathsf{N}$ in (a) matters. It is what makes "the split of $\mathrm{Key}(x)$ at its separators" well defined. Without it, $\mathrm{Key}(x)$ could be written as a join of other words, for example ones containing /, whose prefixes are not Keys.

5 L6: the trie correspondence

L6 says that a model is, up to renaming its entities, nothing more than the set of its name sequences together with a marking of which are items. For example, the model with Keys ε, `d`, `d/e` and `d/f`, where `d/f` is an item, is captured by the four sequences $\langle\rangle$, $\langle\mathtt{d}\rangle$, $\langle\mathtt{d}, \mathtt{e}\rangle$ and $\langle\mathtt{d}, \mathtt{f}\rangle$, with the last one marked. To state L6 we need to say what "the same model" means, and which sets of sequences are allowed.

Definition 5.1 (Isomorphism)**.** [Lean: `Iso`] An *isomorphism* between structures

$$S = (U, F, I, r, C, \mathrm{name}) \quad \text{and} \quad S' = (U', F', I', r', C', \mathrm{name}')$$

is a bijection $f\colon U \to U'$ with $x \in F \iff f(x) \in F'$, $x \in I \iff f(x) \in I'$, $f(r) = r'$, $p \; C \; x \iff f(p) \; C' \; f(x)$, and $\mathrm{name}'(f(x)) = \mathrm{name}(x)$.

Isomorphisms compose and have inverses, and the identity is one.

Definition 5.2 (Tries)**.** [Lean: `Trie, Trie.WF`] A *trie* is a pair (L, K) of sets of finite sequences of words. It is *well formed* when L is a tree of name sequences and K marks some of its leaves. Precisely: every entry of every sequence in L is a name; L contains $\langle\rangle$ and is *prefix closed* ($s \frown t \in L$ implies $s \in L$); and K consists of non-empty sequences in L, none of which has a proper extension in L (if $s \in K$ and $s \frown t \in L$, then $t = \langle\rangle$).

Prefix closure is under dropping whole trailing *names*, not trailing characters. That is why this is a statement about sequences and not about Key strings: in the model with Keys ε, `d`, `d/f`, the string `d/` is a character prefix of a Key and is not a Key.

5.1 From a trie to a model

Definition 5.3. [Lean: `ofTrie`] For a well-formed trie $T = (L, K)$, let S_T be the structure with universe L, items K, folders $L \setminus K$, root $\langle\rangle$, containment $p\ C\ x \iff x = p \frown \langle w \rangle$ for some w, and name(s) the last entry of s (and name($\langle\rangle$) $= \varepsilon$).

Proposition 5.4. [Lean: `ofTrie_axioms`] *S_T is a model of AX-1 to AX-8.*

Proof. The entities are split into K and $L \setminus K$, so each is exactly one of item or folder. The root $\langle\rangle$ is not in K, since K holds only non-empty sequences, so it is a folder.

Containment appends one name. So the only possible container of x is x with its last entry removed, and the empty sequence has no container at all. Every $x = \langle w_1, \ldots, w_n \rangle$ in L is reached from the root through its prefixes $\langle\rangle, \langle w_1 \rangle, \ldots, x$. Each prefix lies in L because L is prefix closed, and each one contains the next.

The root's name is ε, and every other sequence is named by its last entry, which is a name. Two children $p \frown \langle w \rangle$ and $p \frown \langle w' \rangle$ of the same p with the same name have $w = w'$, so they are equal.

Finally, only folders contain anything. If p contains $p \frown \langle w \rangle \in L$, then p has a proper extension in L, so p is not in K. □

Lemma 5.5. [Lean: `seqOf_ofTrie`] *In S_T, $\mathrm{seq}(s) = s$ for every $s \in L$.*

Proof. The prefixes $s{\restriction}0, \ldots, s{\restriction}|s|$ form a chain to s, so by L1 they are the chain of s. The last entry of $s{\restriction}i$ is the i-th entry of s. So by induction on i, the names along the first i steps are $s{\restriction}i$. Take $i = |s|$. □

5.2 From a model to a trie

Definition 5.6. [Lean: `toTrie`] For a model S, let $T_S = (L_S, K_S)$ with $L_S = \{\mathrm{seq}(x) : x \in U\}$ and $K_S = \{\mathrm{seq}(x) : x \in I\}$.

Proposition 5.7. [Lean: `toTrie_wf`] *T_S is well formed.*

Proof. Every sequence in L_S is the name sequence of some entity, so its entries are names, and the root contributes $\langle\rangle$. Prefix closure is Lemma 3.7: the prefix of length i of $\mathrm{seq}(x)$ is the name sequence of x_i, the entity at depth i on the chain of x. Item sequences lie in L_S by definition. They are non-empty: the root is a folder, and no entity is both a folder and an item, so no item is the root. Only the root has the empty sequence (L4).

The one real step is that an item's sequence has no proper extension in L_S. Let x be an item, and suppose $\mathrm{seq}(y)$ properly extends $\mathrm{seq}(x)$. On the chain of y, the entity at depth $|\mathrm{seq}(x)|$ has the same name sequence as x, so it *is* x (Lemma 3.8). It is not the last entity of that chain, so it contains the next one, and only folders contain anything. But x is an item, and no entity is both. □

Proposition 5.8. [Lean: `isoToTrie`] *$x \mapsto \mathrm{seq}(x)$ is an isomorphism from S to S_{T_S}.*

Proof. The map is onto L_S by definition, and one-to-one by Lemma 3.8. Its inverse picks, for each $s \in L_S$, the entity whose sequence is s. It remains to see that it respects each part of the structure.

It sends r to $\langle\rangle$. An entity is an item exactly when its sequence is in K_S, because an entity with the same sequence as an item is that item. Folders are the remaining entities on both sides, so they correspond too. The name of $\mathrm{seq}(x)$ in S_{T_S} is its last entry. That entry is $\mathrm{name}(x)$ by Lemma 3.6, and for the root both names are ε.

If p contains x, then $\mathrm{seq}(x) = \mathrm{seq}(p) \frown \langle \mathrm{name}(x) \rangle$ (Lemma 3.6), so the images are related by containment as well. Conversely, suppose $\mathrm{seq}(x) = \mathrm{seq}(p) \frown \langle w \rangle$. Then x is not the root,

so it has a container q (D1), and $\mathrm{seq}(q) \frown \langle \mathrm{name}(x) \rangle = \mathrm{seq}(p) \frown \langle w \rangle$. Dropping the last entry gives $\mathrm{seq}(q) = \mathrm{seq}(p)$, so $q = p$ and p contains x. □

Lemma 5.9. [Lean: `seqOf_iso`] *If $f : S \to S'$ is an isomorphism of models, then $\mathrm{seq}'(f(x)) = \mathrm{seq}(x)$ for every x.*

Proof. Induction on $\mathrm{depth}(x)$. For $x = r$, $f(r) = r'$ and both sequences are $\langle\rangle$. For $x \neq r$ with container p of smaller depth, $f(p)\ C'\ f(x)$, so $\mathrm{seq}'(f(x)) = \mathrm{seq}'(f(p)) \frown \langle \mathrm{name}'(f(x)) \rangle = \mathrm{seq}(p) \frown \langle \mathrm{name}(x) \rangle = \mathrm{seq}(x)$. □

5.3 The theorem

Theorem 5.10 (L6, trie correspondence)**.** [Lean: `L6`]

(a) For every model S, T_S is a well-formed trie.

(b) For every well-formed trie T, S_T is a model.

(c) For every well-formed trie T, $T_{S_T} = T$.

(d) For every model S, S is isomorphic to S_{T_S}.

(e) Two models S, S' are isomorphic if and only if $T_S = T_{S'}$.

Proof. (a) is Proposition 5.7, (b) Proposition 5.4, and (d) Proposition 5.8.

(c) By Lemma 5.5, $L_{S_T} = \{\mathrm{seq}(s) : s \in L\} = L$ and $K_{S_T} = \{\mathrm{seq}(s) : s \in K\} = K$.

(e) If $f : S \to S'$ is an isomorphism, then by Lemma 5.9 f carries each $\mathrm{seq}(x)$ to the same sequence, and items to items, so $L_S = L_{S'}$ and $K_S = K_{S'}$ (the reverse inclusions use f^{-1}). Conversely, if $T_S = T_{S'}$, compose the isomorphisms $S \to S_{T_S} = S_{T_{S'}} \to S'$ from (d) (`isoOfToTrieEq`). □

So the map $S \mapsto T_S$ is a bijection between isomorphism classes of models and well-formed tries, with inverse $T \mapsto S_T$. The isomorphism preserves the typing (F, I) and the names, not only the containment graph. The prose statement of L6 leaves that open, and the Lean statement fixes it.

Corollary 5.11 (Empty folders)**.** [Lean: `L6_empty_folder`] *In S_T, a leaf of L that is not in K is a folder that contains nothing.*

Proof. It is not in K, so it is a folder. If it contained $x = s \frown \langle w \rangle \in L$, being a leaf would force $\langle w \rangle = \langle\rangle$. □

6 Infinite universes

Spec 051 adds no finiteness axiom on purpose. This section shows that this choice has content in both directions: the axioms admit infinite models, and with infinitely many entities, acyclicity no longer implies reachability.

Theorem 6.1 (An infinite model)**.** [Lean: `infinite_model`] *If Σ has a symbol $a \neq /$, there is a model of AX-1 to AX-8 with an infinite universe.*

Proof. Let L be the set of all finite sequences whose every entry is the one-letter word a, and $K = \emptyset$. This trie is well formed. Its entries are the name a, it contains $\langle\rangle$, and it is prefix closed. The conditions on K hold because K is empty. By Proposition 5.4, $S_{(L,K)}$ is a model. The map $n \mapsto \langle a, \ldots, a \rangle$ (n copies) is injective into its universe, since distinct n give distinct lengths. □

The model is a single infinite downward branch with no items, in which every entity has finite depth. Like the one-point model W_0, it also witnesses that the axioms are *consistent*, so no theorem above holds merely because nothing satisfies the hypotheses.

Theorem 6.2 (Acyclic but not reachable)**.** [Lean: `infinite_upward_chain`] *Let $U = \{r\} \cup \mathbb{N}$, with containment $(n+1)$ C n for every $n \in \mathbb{N}$ and nothing else, every entity a folder, no items, and every name ε. Then every non-root has a container, r has none, C has no cycle, and AX-6 fails.*

Proof. n is contained by $n+1$, and nothing contains r. Along any C-path $y_0, \ldots, y_k$ with $k \geq 1$, every y_i is a number and $y_0 = y_i + i$ (by induction on i, as each step goes from $m+1$ to m). So $y_k = y_0$ forces $k = 0$. Finally, 0 has no chain. A chain of length 0 would make $0 = r$, and a chain of length ≥ 1 would start with $r\ C\ x_1$, but r contains nothing. □

All names here are ε, so this structure also violates AX-7. It shows why AX-6 cannot be replaced by acyclicity; it is not an independence witness for AX-6 (that is row AX-6 of Table 1).

Proposition 6.3 (The finite half)**.** *If U is finite, every non-root has a container, r has none, and C has no cycle, then AX-6 holds.*

Proof. Given x, choose a container of x (if $x \neq r$), then a container of that, and so on, as long as the current entity is not r. Two equal entities in this walk would give a cycle, so its entries are distinct, and as U is finite the walk stops. It can stop only at an entity without a container, that is at r. Reversed, the walk is a chain to x. □

This proposition is *not* in the Lean file. It is checked in Alloy for universes of up to six entities (`finite_acyclic_gives_reach`). It explains why, in a finite universe, acyclicity (with every non-root contained and r uncontained) already implies AX-6, so a bounded model finder cannot separate the two. The reason the axioms keep AX-6 is Theorem 6.2: without finiteness, acyclicity is not enough. The other direction needs AX-4: the structure in the remark after L2 is finite, satisfies AX-6, and has a cycle.

7 What the bounded Alloy model adds

The Lean file proves consequences *of* the axioms for every universe. It does not show that each axiom clause is needed. That is a question about structures that *violate* the axioms, and a model finder answers it cheaply. The Alloy model restates the eleven clauses as eleven predicates and runs 30 commands, each with an expected outcome (a `run` must find an instance; a `check` must find no counterexample). `scripts/alloy_check.sh` fails if any outcome differs.

7.1 Independence

For each clause, Table 1 gives a finite structure that satisfies every other clause and violates that one. Unless a row says otherwise, $\text{name}(r) = \varepsilon$ and the other entities have distinct names in `N`. Each row is a finite object, so checking it by hand is direct inspection. Alloy confirms each one with a `run` at scope 4 (entities) and sequence length 6. For the rows whose damage is a claim about Keys (AX-7 non-empty, AX-7 no-/, AX-8), a `check` confirms that two entities get the same Key. For AX-4, a `run` exhibits two different chains to x.

The AX-7 *root* row deserves a word. Nothing in the definition of Key reads $\text{name}(r)$, so no lemma fails without this clause. It is a normalization, and it matters only for L6, where $\text{name}(r)$ would otherwise be free data that the trie does not record.

Clause broken	Structure	What goes wrong
AX-1 disjoint	$U = \{r, x\}$, $F = \{r, x\}$, $I = \{x\}$, $C = \{(r, x)\}$	x is both folder and item.
AX-1 covering	$U = \{r, x\}$, $F = \{r\}$, $I = \emptyset$, $C = \{(r, x)\}$	x has no kind.
AX-2	$U = \{r\}$, $F = \emptyset$, $I = \{r\}$, $C = \emptyset$	The root is an item.
AX-3	$U = \{r, x, y\}$, $F = \{r\}$, $I = \{x, y\}$, $C = \{(r, x), (x, y)\}$	An item contains something.
AX-4	$U = \{r, a, b, x\}$, $F = \{r, a, b\}$, $I = \{x\}$, $C = \{(r, a), (r, b), (a, x), (b, x)\}$	Two chains, so two Keys, for x.
AX-5	$U = F = \{r, p\}$, $C = \{(r, p), (p, r)\}$	A cycle through r.
AX-6	$U = F = \{r, g_1, g_2\}$, $C = \{(g_1, g_2), (g_2, g_1)\}$	A detached cycle.
AX-7 root	$U = F = \{r\}$, $C = \emptyset$, name$(r) =$ `z`	Only a normalization breaks. No Key changes.
AX-7 non-empty	$U = \{r, x\}$, $I = \{x\}$, $C = \{(r, x)\}$, name$(x) = \varepsilon$	Key$(x) = \varepsilon =$ Key(r).
AX-7 no /	$U = F = \{r, w, a, b\}$, $C = \{(r, w), (r, a), (a, b)\}$, name$(w) =$ `a/b`, name$(a) =$ `a`, name$(b) =$ `b`	Key$(w) =$ Key$(b) =$ `a/b`.
AX-8	$U = \{r, x, y\}$, $I = \{x, y\}$, $C = \{(r, x), (r, y)\}$, name$(x) =$ name(y)	Key$(x) =$ Key(y).

Table 1: One countermodel per axiom clause (spec 051's rationale, §5). Where I is omitted, $I = \emptyset$; where F is omitted, $F = U \setminus I$.

7.2 Consistency and witnesses

The witness models W_0 (the root alone), W_1 (Keys ε, `d`, `d/f`) and W_2 (Keys ε, `d`, `d/e`, `d/f`, with `e` an empty folder) satisfy all eight axioms, and their Keys are the ones listed. Together with Theorem 6.1, they show that the axioms are consistent, in finite and in infinite universes.

7.3 The lemmas, in bounded scope

D1 is checked at scope 6. L1 to L5 are checked at scope 5. The Key lemmas L3 to L5 are checked with names of at most two symbols, while L1 and L2 have no separate bound on name length. The sequence scope is chosen so that no structure in scope is excluded by the encoding of Keys. With at most five entities the depth is at most 4, and a Key has at most one name and one / per level and no trailing /, so its length is at most $4 \cdot 3 - 1 = 11$. A separate `run` shows that length 11 is actually reached.

Alloy cannot state L6. L6 quantifies over pairs of structures and over isomorphisms between them, and an Alloy instance is a single structure. It also cannot reach Theorems 6.1 and 6.2. The checks are therefore evidence for finite universes up to the stated scope, and the Lean proofs are what extend them to all universes.

7.4 The checks are not vacuous

A check that passes because the axioms are too strong proves nothing. The model was mutated by deleting one conjunct at a time from the axioms (mutants run once and not committed). Without AX-4, D1, L1 and L2 fail. Without AX-8, L3 fails. Without AX-3, L5 fails. Without AX-7 (no /), L3, L4 and L5 fail. This matches the proofs above. AX-4 enters directly in D1

and through Lemma 3.1, AX-8 through Lemma 3.8, AX-3 through Theorem 4.7, and the no-/ clause through Lemma 2.4 (L3, L5) and Lemma 2.6 (L4).

7.5 Division of labour

Question	Alloy (bounded)	Lean (all universes)
Each clause independent	yes, 11 rows	—
Axioms consistent	W_0, W_1, W_2	infinite model, Thm. 6.1
D1, L1 to L5	up to scope 5–6	proved
L6	cannot be stated	proved
Finite half of §6	up to scope 6	—
Infinite upward chain	out of reach	proved

A Concordance with the sources

Here	Spec 051	Lean declaration
AX-1 to AX-8	TREE-001 to TREE-008	`Axioms` (fields `ax1_disjoint` … `ax8`)
Lemma 2.2	—	`split_first`
Lemma 2.4	—	`join_inj`
Lemma 2.6	—	`join_shape`
Lemma 3.1	—	`chain_unique`
Lemma 3.3	—	`chain_step`
Theorem 3.4 (D1)	—	`D1`
Definition 3.5 (Key)	TREE-009	`Key`
Lemma 3.8	—	`seqOf_inj`
L1, L2	—	`L1`, `L2`
L3	TREE-010	`L3`
L4	TREE-011	`L4`
L5	TREE-012	`L5`
Theorem 5.10 (L6)	—	`L6`, `L6_empty_folder`
Theorem 6.1	—	`infinite_model`
Theorem 6.2	—	`infinite_upward_chain`
Proposition 6.3	—	(Alloy only)

The Lean file uses only Lean's core library and its standard axioms (propositional extensionality, choice, quotient soundness), never `sorry`. A `#guard_msgs` block pins the axioms each headline result depends on. Choice enters where this text says "the chain of x" and "the entity whose sequence is s", and in the excluded middle used for AX-1 (covering) of S_T (Proposition 5.4). The last is why Theorem 6.1 depends on choice although it uses no chain.